1. Student Result Analysis Project with Python & Data Analysis

1.0.1 Import Liabraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

1.0.2 Importing the Data

```
df = pd.read_csv("Student Result Analysis Data.csv")
```

```
df.head()
```

Columns 1–8 of 16

	Unnamed: 0	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus
0	0	female	NaN	bachelor's degree	standard	none	married
1	1	female	group C	some college	standard	NaN	married
2	2	female	group B	master's degree	standard	none	single
3	3	male	group A	associate's degree	free/reduced	none	married
4	4	male	group C	some college	standard	none	married

Columns 9–15 of 16 · first column repeated

	PracticeSport	IsFirstChild	NrSiblings	TransportMeans	WklyStudyHours	MathScore	ReadingScore
0	regularly	yes	3.0	school_bus	< 5	71	71
1	sometimes	yes	0.0	NaN	10-May	69	90
2	sometimes	yes	4.0	school_bus	< 5	87	93
3	never	no	1.0	NaN	10-May	45	56
4	sometimes	yes	0.0	school_bus	10-May	76	78

Columns 16–16 of 16 · first column repeated

	WritingScore
0	74
1	88
2	91
3	42
4	75

```
df.describe()
```

	Unnamed: 0	NrSiblings	MathScore	ReadingScore	WritingScore
count	30641.000000	29069.000000	30641.000000	30641.000000	30641.000000
mean	499.556607	2.145894	66.558402	69.377533	68.418622
std	288.747894	1.458242	15.361616	14.758952	15.443525
min	0.000000	0.000000	0.000000	10.000000	4.000000
25%	249.000000	1.000000	56.000000	59.000000	58.000000
50%	500.000000	2.000000	67.000000	70.000000	69.000000
75%	750.000000	3.000000	78.000000	80.000000	79.000000
max	999.000000	7.000000	100.000000	100.000000	100.000000

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 30641 entries, 0 to 30640
Data columns (total 15 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   Unnamed: 0           30641 non-null  int64  
 1   Gender               30641 non-null  str    
 2   EthnicGroup          28801 non-null  str    
 3   ParentEduc           28796 non-null  str    
 4   LunchType            30641 non-null  str    
 5   TestPrep             28811 non-null  str    
 6   ParentMaritalStatus  29451 non-null  str    
 7   PracticeSport        30010 non-null  str    
 8   IsFirstChild         29737 non-null  str    
 9   NrSiblings           29069 non-null  float64
 10  TransportMeans       27507 non-null  str    
 11  WklyStudyHours       29686 non-null  str    
 12  MathScore            30641 non-null  int64  
 13  ReadingScore         30641 non-null  int64  
 14  WritingScore         30641 non-null  int64  
dtypes: float64(1), int64(4), str(10)
memory usage: 5.6 MB
```

```
df.isnull().sum()
```

```
Unnamed: 0               0
Gender                   0
EthnicGroup           1840
ParentEduc            1845
LunchType                0
TestPrep              1830
ParentMaritalStatus   1190
PracticeSport          631
IsFirstChild           904
NrSiblings            1572
TransportMeans        3134
WklyStudyHours         955
MathScore                0
ReadingScore             0
WritingScore             0
dtype: int64
```

1.0.3 Drop Unnamed Column

```
df = df.drop("Unnamed: 0", axis = 1)
print(df.head())
```

```
   Gender EthnicGroup          ParentEduc     LunchType TestPrep  \
0  female         NaN   bachelor's degree      standard     none
1  female     group C        some college      standard      NaN
2  female     group B     master's degree      standard     none
3    male     group A  associate's degree  free/reduced     none
4    male     group C        some college      standard     none

  ParentMaritalStatus PracticeSport IsFirstChild  NrSiblings TransportMeans  \
0             married     regularly          yes         3.0     school_bus
1             married     sometimes          yes         0.0            NaN
2              single     sometimes          yes         4.0     school_bus
3             married         never           no         1.0            NaN
4             married     sometimes          yes         0.0     school_bus

  WklyStudyHours  MathScore  ReadingScore  WritingScore
0            < 5         71            71            74
1         10-May         69            90            88
2            < 5         87            93            91
3         10-May         45            56            42
4         10-May         76            78            75
```

1.0.4 Change Weekly Study Hours Column

```
df['WklyStudyHours'] = df['WklyStudyHours'].str.replace("10-May", "5-10")
df.head()
```

Columns 1–8 of 15

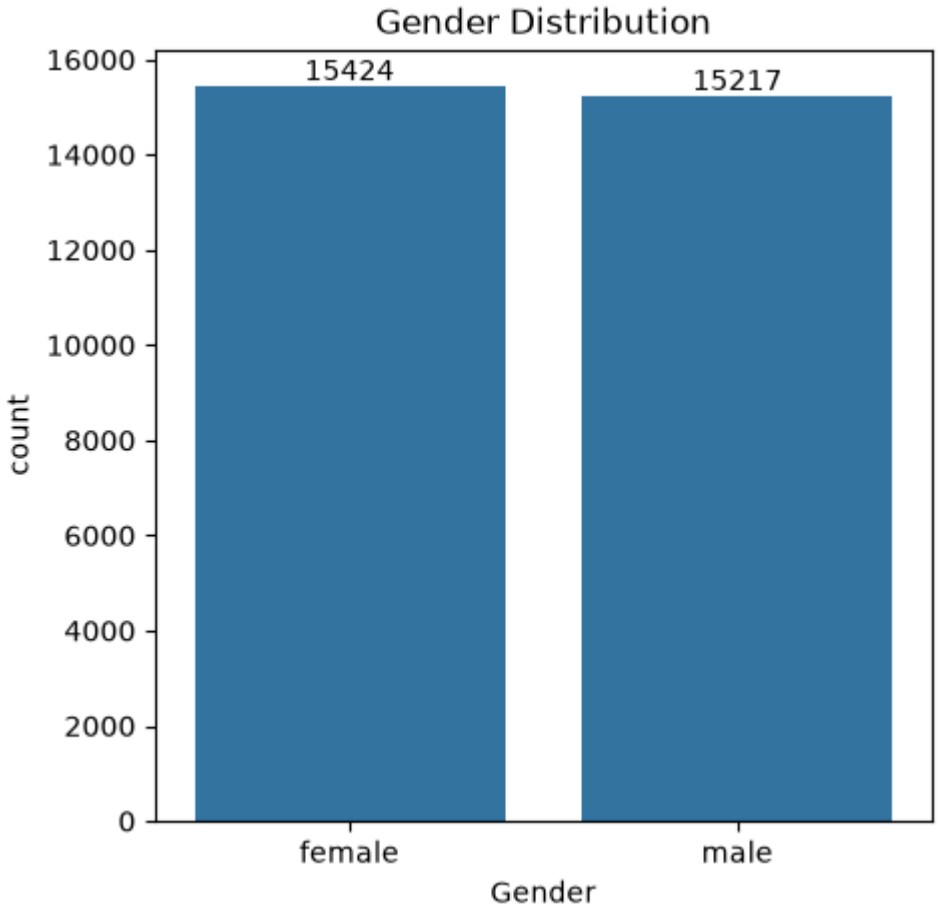
	Gender	EthnicGroup	ParentEduc	LunchType	TestPrep	ParentMaritalStatus	PracticeSport
0	female	NaN	bachelor's degree	standard	none	married	regularly
1	female	group C	some college	standard	NaN	married	sometimes
2	female	group B	master's degree	standard	none	single	sometimes
3	male	group A	associate's degree	free/reduced	none	married	never
4	male	group C	some college	standard	none	married	sometimes

Columns 9–15 of 15 · first column repeated

	IsFirstChild	NrSiblings	TransportMeans	WklyStudyHours	MathScore	ReadingScore	WritingScore
0	yes	3.0	school_bus	< 5	71	71	74
1	yes	0.0	NaN	5-10	69	90	88
2	yes	4.0	school_bus	< 5	87	93	91
3	no	1.0	NaN	5-10	45	56	42
4	yes	0.0	school_bus	5-10	76	78	75

1.0.5 Gender Dristribution

```
plt.figure(figsize = (5,5))
ax = sns.countplot(data = df, x = "Gender")
ax.bar_label(ax.containers[0])
plt.title("Gender Distribution")
plt.show()
```

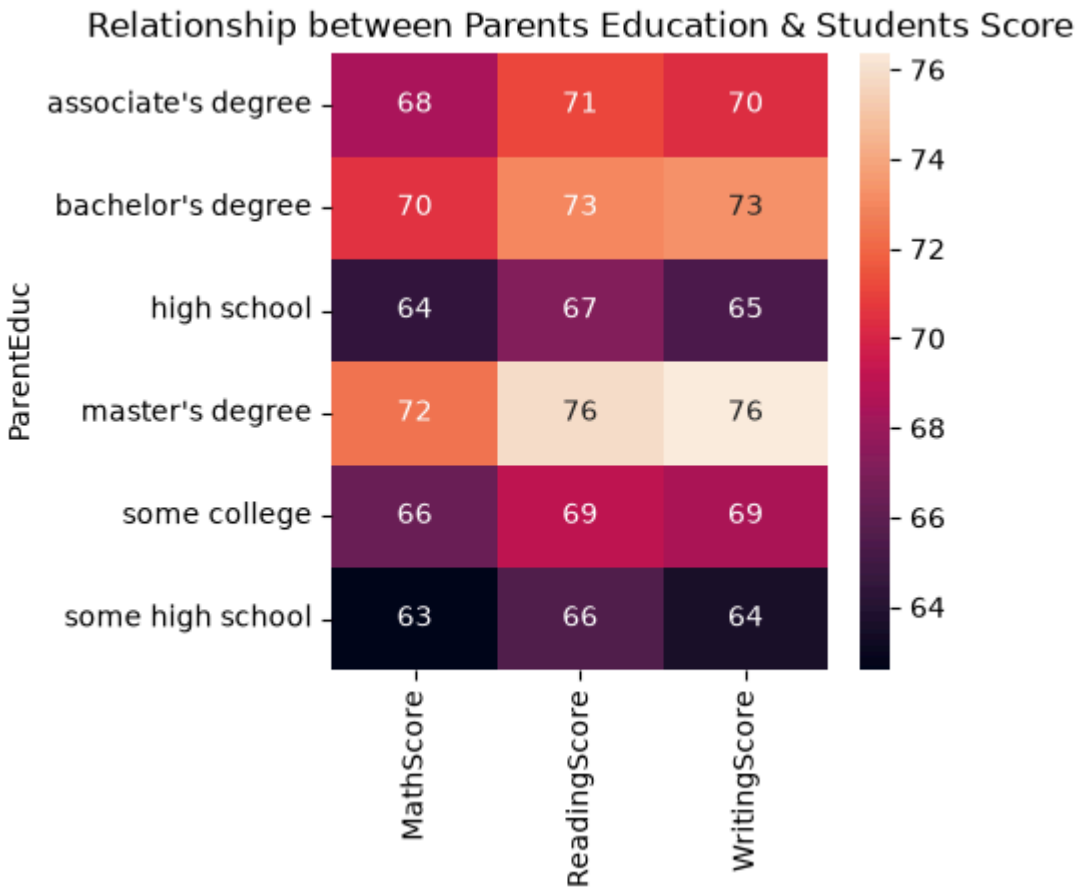


From the above chart we have analysed that the number of females in the data is more than the number of males

```
gb = df.groupby("ParentEduc").agg({"MathScore" : 'mean', "ReadingScore" : 'mean',
"WritingScore" : 'mean'})
```

```
plt.figure(figsize = (4,4))
sns.heatmap(gb, annot = True)
plt.title("Relationship between Parents Education & Students Score")
```

```
Text(0.5, 1.0, 'Relationship between Parents Education & Students Score')
```

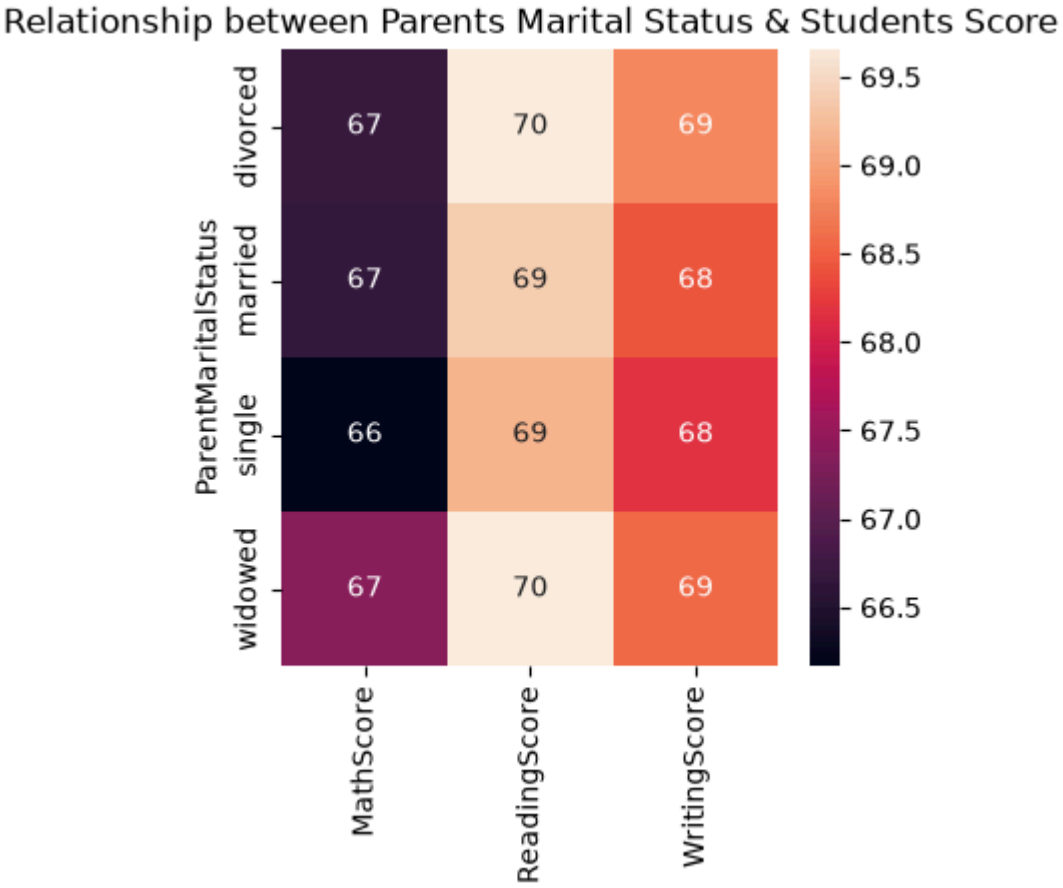


From the above chart we have concluded that the Parents education have a good impact on Students score

```
gb1 = df.groupby("ParentMaritalStatus").agg({"MathScore" : 'mean', "ReadingScore" :
'mean', "WritingScore" : 'mean'})
```

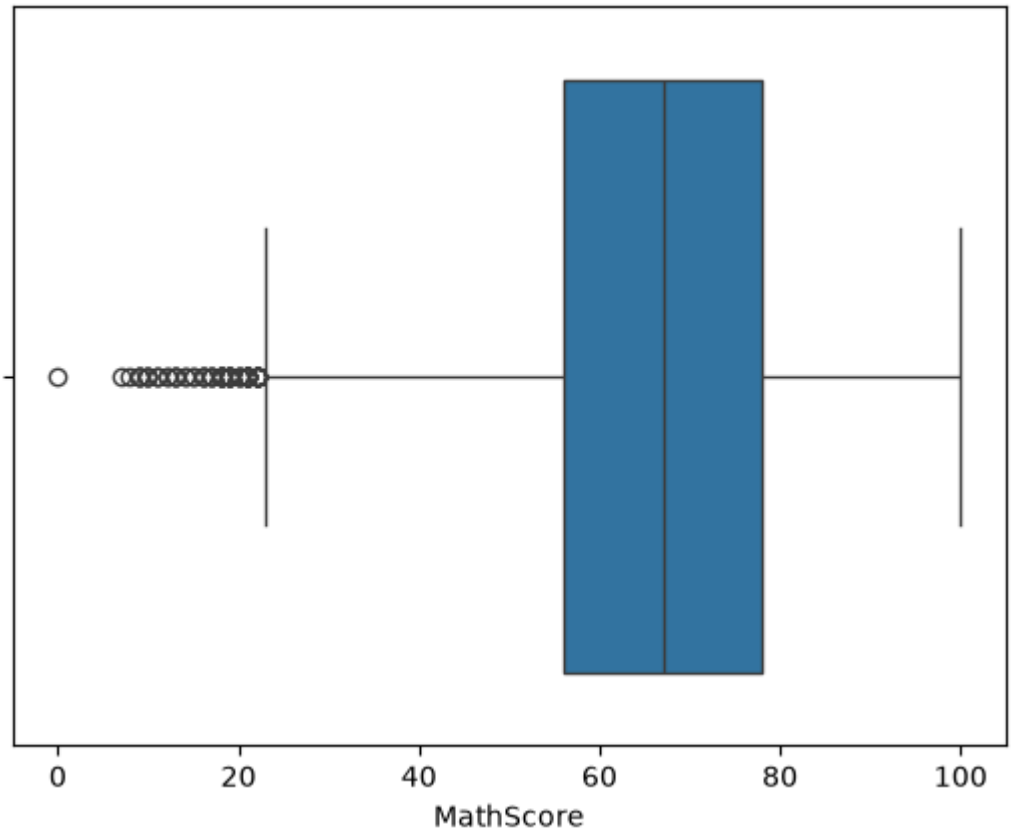
```
plt.figure(figsize = (4,4))
sns.heatmap(gb1, annot = True)
plt.title("Relationship between Parents Marital Status & Students Score")
```

```
Text(0.5, 1.0, 'Relationship between Parents Marital Status & Students Score')
```

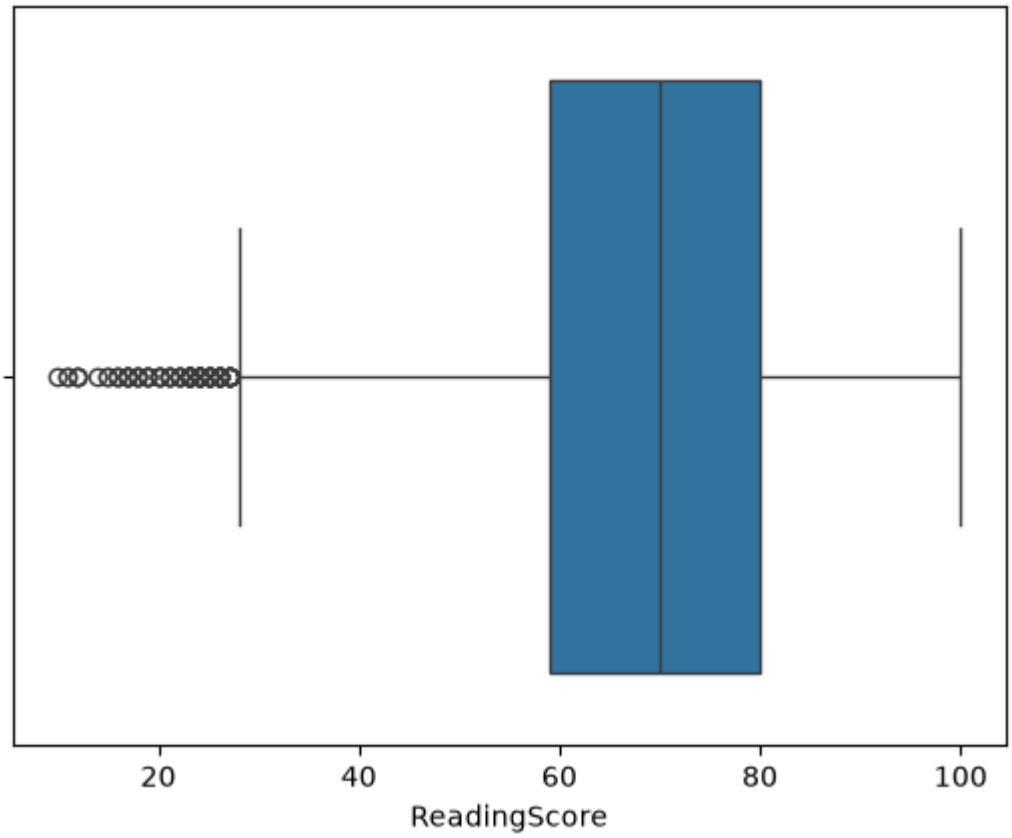


From the above chart we have concluded that there is no / negligible impact on the students score due to theier parents marital status.

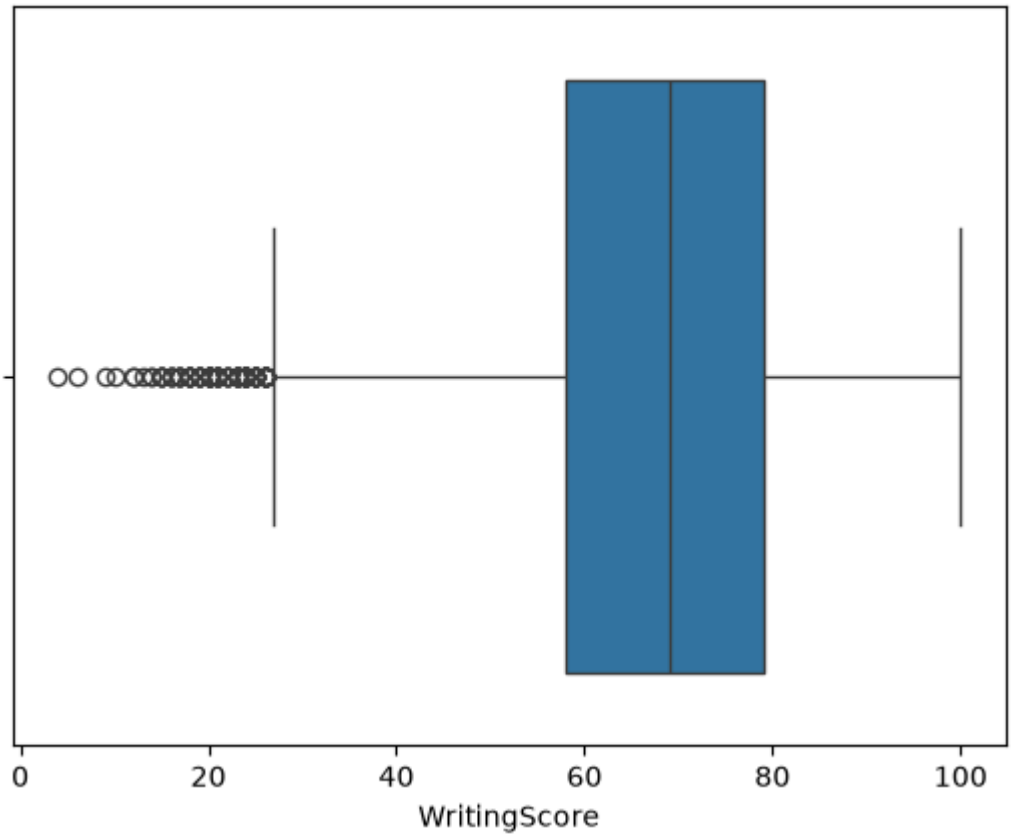
```
sns.boxplot(data = df, x = "MathScore")
plt.show()
```



```
sns.boxplot(data = df, x = "ReadingScore")
plt.show()
```



```
sns.boxplot(data = df, x = "WritingScore")
plt.show()
```




```
print(df['EthnicGroup'].unique())
```

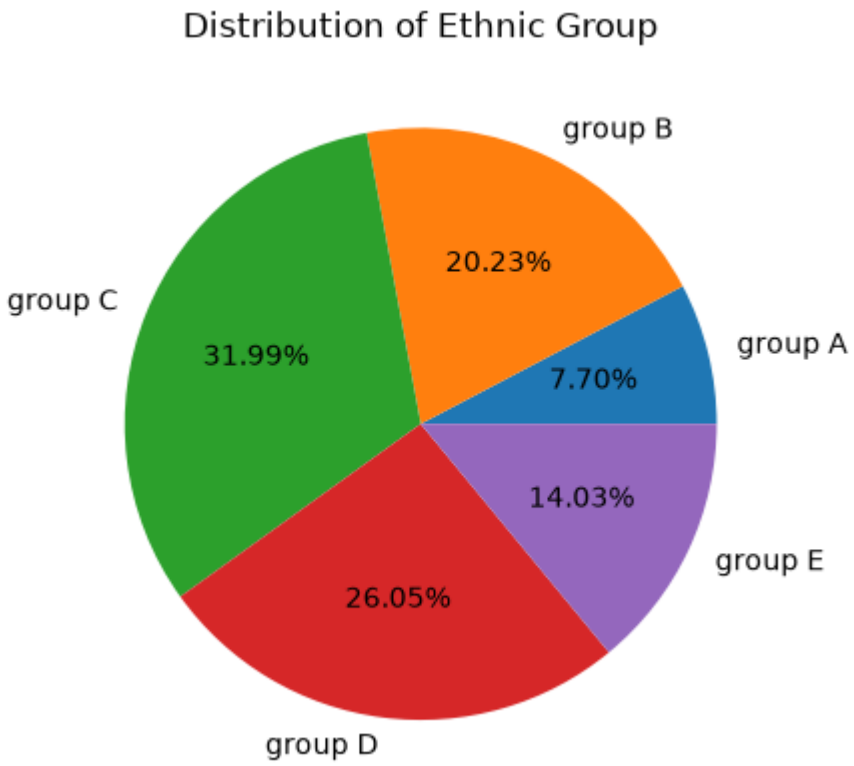
```
<ArrowStringArray>
[nan, 'group C', 'group B', 'group A', 'group D', 'group E']
Length: 6, dtype: str
```

1.0.6 Distribution of Ethnic Groups

```
groupA = df.loc[(df['EthnicGroup'] == "group A")].count()
groupB = df.loc[(df['EthnicGroup'] == "group B")].count()
groupC = df.loc[(df['EthnicGroup'] == "group C")].count()
groupD = df.loc[(df['EthnicGroup'] == "group D")].count()
groupE = df.loc[(df['EthnicGroup'] == "group E")].count()

l = ["group A", "group B", "group C", "group D", "group E" ]
mlist = (groupA['EthnicGroup'], groupB['EthnicGroup'], groupC['EthnicGroup'],
groupD['EthnicGroup'], groupE['EthnicGroup'])

plt.pie(mlist, labels = l, autopct = "%1.2f%%")
plt.title("Distribution of Ethnic Group")
plt.show()
```



```
ax = sns.countplot(data = df, x = 'EthnicGroup' )
ax.bar_label(ax.containers[0])
```

```
[Text(0, 0, '9212'),
 Text(0, 0, '5826'),
 Text(0, 0, '2219'),
 Text(0, 0, '7503'),
 Text(0, 0, '4041')]
```

